

Chapter: 25

QUANTUM PHYSICS

QUANTUM PHYSICS:

Introduction:

In the early 20th century, the field of quantum physics was developed because of certain experimental discoveries which could not be described by classical physics.

The fundamental discovery of quantum physics was that:

Energy is radiated as discrete packets, or quanta.

It aims to discover the properties and behaviors of the very building blocks of nature.

Definition:

"The study of the behavior of matter and energy at atomic and sub-atomic levels is called Quantum Physics".

Explanation:

Quantum physics explains how very tiny particles (like electrons and photons) behave in discrete packets of energy (quanta) and can show both wave and particle nature.

At atomic and sub-atomic levels, the classical laws of physics no longer apply, and strange, seemingly random phenomena govern the behavior of particles.

Quantum physics explores the principles of wave-particle duality, uncertainty, superposition and quantization, which form the basis of our understanding of the behavior of matter, energy, and the fabric of reality itself.

QUANTUM THEORY OF RADIATION:

Quantum theory of radiation was introduced by Max Planck and Albert Einstein.

Statement:

Energy is emitted or absorbed in small, discrete packets called quanta, and not in a continuous manner.

Each quantum of radiation has energy given by

$$E = h\nu$$

Where h is Planck's constant and ν is the frequency of radiation.

BLACK BODY RADIATION:

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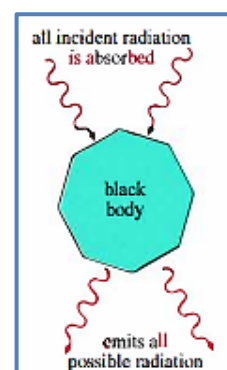
Definition:

A black body is an idealized object which absorbs all incident radiation and also emits it in a continuous spectrum of colors, depending upon its temperature.

Explanation:

When a body is heated, it gives out energy in the form of radiation. The type of radiation it emits depends on how hot the body is.

- At low temperatures, the body emits radiation with long wavelengths, mainly in the infrared region, which we cannot see.



- As the temperature increases, the body starts emitting shorter wavelengths, and the amount of radiation is not the same for all wavelengths.

For a long time (until the 19th century), scientists could not explain the pattern of light emitted by heated objects.

- At normal room temperature, objects emit electromagnetic radiation, but it is very weak, so we cannot see or feel it.
- When the temperature increases, objects start to emit infrared radiation, which we feel as heat when we go near them.
- If the object becomes very hot, it begins to glow:
 - At lower high temperatures, it glows red (red-hot).
 - At temperatures above 2000 K, it glows yellow or white, like white-hot iron.

The light given out by a hot object contains all wavelengths, not just one color. This is called a continuous spectrum.

A spectrum graph shows how strong the light is (intensity) at different wavelengths or frequencies.

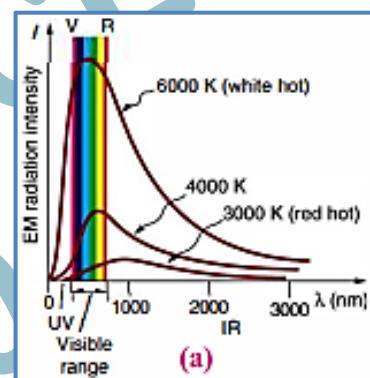
As the temperature increases:

- The total energy of radiation increases.
- The peak of the radiation moves toward shorter wavelengths (from infrared to visible and then ultraviolet light).

Scientists studied this radiation using an ideal object called a blackbody.

- The intensity of radiation increases with temperature.
- The peak of the spectrum shifts toward visible and ultraviolet regions as temperature rises.

Classical physics could not explain the exact shape of this spectrum, which later led to the development of quantum physics.



SOME EXPERIMENTAL FACTS ABOUT BLACKBODY RADIATIONS:

1. It depends only on temperature, not material:

The radiation given out by a hot object depends **only on its temperature**, not on what it is made of.

For example: an iron horseshoe, a ceramic vase, and charcoal will all give the same radiation if they are at the same temperature.

When the temperature increases, the object gives out more radiation at all wavelengths.

2. Peak shifts to shorter wavelengths when temperature increases

As an object becomes **hotter**, the brightest part of its radiation moves toward shorter wavelengths (called blue shift).

For example: blue stars are hotter than red stars.

3. Less radiation at very short wavelengths:

Even though the peak shifts to shorter wavelengths at high temperature, the radiation at the extreme short-wavelength (high-frequency) side is always small. The curve drops off on the left side of the spectrum.

LAWS OF BLACK BODY RADIATION:

1. RAYLEIGH-JEANS LAW:

Statement:

"The energy per unit volume per unit wavelength (Intensity) of a blackbody radiation is inversely proportional to the fourth power of the wavelength (λ)".

Mathematically:

This relationship is mathematically expressed as:

$$E(\lambda, T) = \frac{\text{constant}}{\lambda^4}$$

Where “ $E(\lambda, T)$ ” is the spectral radiance, λ is the wavelength,

2. WIEN'S DISPLACEMENT LAW:**Statement:**

The wavelength at which the intensity of blackbody radiation is maximum is inversely proportional to the temperature of the blackbody. OR

“Wavelength of maximum intensity λ_m is inversely proportional to the temperature of a black body radiator.”

As the temperature increases, the peak of the blackbody radiation curve shifts to shorter wavelengths.

Mathematically:

This relationship is mathematically expressed as:

$$\begin{aligned}\lambda_{\max} &\propto \frac{1}{T} \\ \lambda_{\max} &= \frac{\text{constant}}{T} \\ \lambda_{\max} \times T &= \text{Constant}\end{aligned}$$

Where $\lambda_{\max}$ is the wavelength at which a blackbody radiates at a given temperature T.

3. STEFAN'S LAW:**Statement:**

“The total energy radiated per unit time per unit surface area of a black body is directly proportional to the fourth power of its absolute temperature”.

Mathematically:

Mathematically, it is expressed as:

$$\begin{aligned}\frac{E}{t} \cdot \frac{1}{A} &\propto T^4 \\ P \cdot \frac{1}{A} &= \sigma T^4 \\ P &= \sigma AT^4\end{aligned}$$

Where A is the surface area of a blackbody, T is its temperature and σ is the Stefan's-Boltzmann constant, $\sigma = 5.67 \times 10^{-8} \text{ W / (m}^2 \cdot \text{K}^4\text{)}$.

PLANCK'S HYPOTHESIS OR ASSUMPTION:

Planck explained blackbody radiation by introducing the idea of quantized energy.

Statement:

According to Planck's Hypothesis:

The energy levels of the oscillators (tiny vibrating particles) in the walls of a blackbody are quantized, meaning they can only have specific, discrete values.

Explanation:

Planck proposed that:

- Energy is not continuous; it exists in small packets called quanta.
- Electromagnetic energy is emitted or absorbed only in “quanta”.

- Each quantum of energy is equal to hf .

Where h = Planck's constant, f = frequency of radiation

Energy of Oscillator:

The energy (E_n) of an oscillator in the cavity walls is given by the equation:

$$E_n = nhf$$

Where ($n = 1, 2, 3, \dots$), h is Planck's constant, f is the frequency of the oscillator.

This relationship, now known as Planck's Law, marked the birth of quantum theory, fundamentally changing our understanding of energy and leading to the development of quantum mechanics

Absorption and Emission:

An oscillator in the wall can receive energy from the radiation in the cavity (absorption), or it can emit energy to the radiation in the cavity (emission).

The absorption process sends the oscillator to a higher quantum state, and the emission process sends the oscillator to a lower quantum state.

Whichever way this exchange of energy goes, the smallest amount of energy allowed to exchange is hf .

There is no upper limit to how much energy can be exchanged, but whatever is exchanged must be an integer multiple of hf .

If the energy packet does not have this exact amount, it is neither absorbed nor emitted on the wall of the blackbody.

SELF-ASSESSMENT QUESTIONS:

- How does the energy distribution of black body radiation change with temperature? Describe the relationship between temperature and the peak wavelength of radiation.
- Describe the ultraviolet catastrophe problem in the context of black body radiation. How did Max Planck's quantum hypothesis resolve this issue?

WORKED EXAMPLE:

25.1: At the Earth's surface, the energy flux in sunlight is $1.02 \times 10^3 \text{ watt/m}^2$. If a black sheet of paper faces the Sun, what is the equilibrium temperature of the paper? Assume that the bottom of the paper is insulated, so the only heat loss is by blackbody radiation from the top surface.

PARTICLE NATURE OF ELECTROMAGNETIC RADIATION:

Electromagnetic radiation can be described in terms of a stream of massless particles, called photons.

Each photon contains a certain amount of energy. The different types of radiation are defined by the amount of energy found in the photons.

The particle nature of light states that:

Light consists of particles called 'Photons' (Particles do not interfere) That is, when the space is occupied by some particles, other particles cannot occupy the same space.

Experiments from the last century, such as the **photoelectric effect** and **atomic line spectra**, can only be explained if EM radiation is assumed to behave as particles.

Energy of Photon in wave nature:

The energy of a photon depends upon the frequency as described in Planck's hypothesis.

$$E = h\nu$$

But $\nu = \frac{c}{\lambda}$. Hence by putting the value in above equation, we get

$$E = \frac{hc}{\lambda}$$

Where: h = Planck's constant = 6.626×10^{-34} , c = Speed of light = $3 \times 10^8 \text{ m/s}$, λ = Wavelength of Photon particles

WORKED EXAMPLE:

25.2: Calculate the energy of a photon with a frequency of 3×10^{18} Hz.

PHOTOELECTRIC EFFECT:**Definition:**

"The emission of electrons from a metal surface when exposed to light of suitable frequency is called the photoelectric effect."

When electromagnetic radiation like light is shined on certain metallic materials, electrons are emitted due to absorption of light by the electrons on the surface of material.

Explanation:

In 1887, Heinrich Hertz observed the photoelectric effect in Germany, discovering that:

Ultraviolet (UV) light shining onto metal surfaces causes the ejection of electrons, which are pushed with potential to reach an anode.

In 1899, J.J. Thomson in England demonstrated that:

The incidence of UV light on a metal surface causes the ejection of electrons.

In 1902, Philip Lenard measured the photoelectric effect quantitatively and concluded that:

Light has electrical properties.

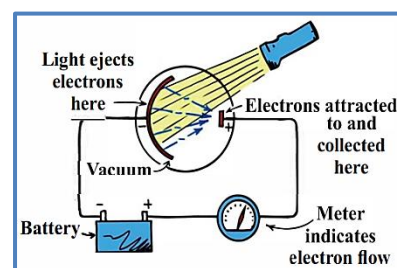
Though these experiments were able to describe phenomenon of photoelectric effects, some situations during experiments were not answered with scientific logic.

In 1905, Albert Einstein was able to resolve the phenomenon.

EXPERIMENTAL EXPLANATION:**ARRANGEMENT OF APPARATUS:**

The photoelectric phenomenon is shown in figure, in which:

- An evacuated tube containing two electrodes cathode and anode; connected with a variable source of voltage V .
- A monochromatic light source is shined onto cathode through a quartz window. and anode (collecting electrode) is connected either positive or negative potential with respect to the cathode.
- An ammeter is connected with circuits to record the current due to photoelectrons.

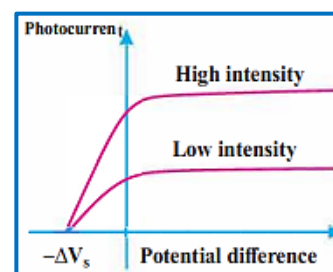
**OBSERVATIONS:**

1. For a constant potential difference between the cathode and anode, the number of electrons emitted from cathode increases with increasing intensity of radiation.
2. For a constant intensity and frequency of incident radiation the photoelectric current varies with the potential difference V between the cathode and anode and reaches a constant value beyond which further increase of potential difference does not affect the photoelectric current, instead, if the anode is made more and more negative with respect to the photo cathode surface the current decreases.

This negative potential (with respect to cathode) of the plate is called retarding potential.

As shown in the graph:

For a particular value of retarding potential, the photoelectric current becomes zero. This potential is called cut-off or stopping potential.



Cut-off or stopping potential V_o is measure of maximum kinetic energy of photoelectrons.

We can write $K.E_{max} = eV_o$

Where $K.E_{max}$ (the maximum kinetic energy of the ejected electron.)

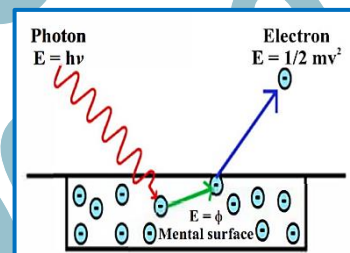
- The stopping potential and hence the maximum kinetic energy $K.E_{max}$ of photoelectrons is independent of the intensity of incident radiation and depends only on the frequency " ν " of radiation.
- For each substance, there exists a characteristic frequency ν_o such that radiation with frequency below the photoelectrons is not ejected from the surface. This frequency is called the threshold frequency, and the corresponding wavelength is called Threshold wavelength, λ_o ($\frac{hc}{\phi}$)
- The time lag between the incidence of radiation and the emission of a photoelectron is very small, less than 10^{-9} seconds.

EXPLANATION OF PHOTO-ELECTRIC EFFECT ON THE BASIS OF QUANTUM THEORY:

Einstein proposed that radiation energy is not continuously distributed over the wavefront, but the light energy consists of discrete quanta of energy $h\nu$, which penetrates the surface of the cathode, all of its energy is transformed into an electron; it's shown in figure.

The photon's energy is associated with its frequency ν , through a proportionality constant known as Planck's constant h , or alternately, using the wavelength λ and the speed of light c .

$$\begin{aligned} K.E &= h\nu \\ &= \frac{hc}{\lambda} \\ \text{Or the momentum equation: } p &= \frac{h}{\lambda} \end{aligned}$$



EINSTEIN'S EXPLANATION:

According to Einstein, a beam of light of frequency " ν " consists of a stream of energy corpuscles called photons; the energy of each photon being " $h\nu$ " a photon could be absorbed by a single electron in the metal surface.

The electron needs a certain minimum energy called the work function ϕ to escape from the metal surface.

Now if the energy of incident photons is greater than or at least equal to the maximum energy (or work function ϕ), the electron is ejected instantaneously from the metal surface.

EINSTEIN'S PHOTOELECTRIC EQUATION:

When a photon is incident on the surface of a material, some of its energy is spent in making the electron free and the rest appears as kinetic energy of the electron.

The electrons at the surface of the material are most loosely bound and require minimum energy for their liberation. This energy is called the work function ϕ of the material.

If a photon of energy " $h\nu$ " greater than the work function ($h\nu > \phi$) of the metal surface falls upon it, the photon transfers its whole energy to the electron of the metal.

A part of this energy equal to work function ϕ is used in bringing the electron out of the metal surface (i.e. break away from the surface) and the remaining energy appears as the kinetic energy of the electron. That is.

Incident photon energy - work function = Max K.E of photoelectrons.

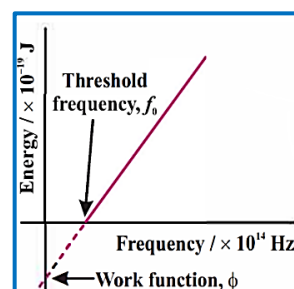
Or $h\nu - \phi = K.E_{max}$

The maximum kinetic energy of photo electrons ejected from the surface is given by

$$K.E_{max} = \frac{1}{2}mv^2 = h\nu - \phi$$

This relation is known as Einstein's photoelectric equation.

So, work function is the minimum energy that must be supplied to the electron for it to leave the Metal surface. As shown in figure.



Consequently, there exists a minimum frequency that is independent of the intensity of light, below which electrons cannot be ejected from the metal.

This proposal was proved by Millikan when in 1914, he published his results of the voltage required to stop photoelectrons ejecting from a metal surface.

SELF-ASSESSMENT QUESTIONS:

1. How does the photoelectric effect demonstrate the particle nature of light?
2. Calculate the threshold frequency of incident radiation that will cause the emission of photo electrons from the surface of gold.

WORKED EXAMPLE:

25.3: The work function for zinc is 4.24 eV. What is the threshold frequency for the ejection of photoelectrons from zinc?

PHOTOELECTRIC EFFECT IN SOLAR CELLS AND PHOTOCELLS:

Photovoltaic cells are made up of p-type and n-type silicon semiconductors. When light falls on the junction between p-type and n-type semiconductors, electrons are emitted according to the photoelectric effect. These electrons are collected in an external circuit to produce an electric current.

- Photocell, a device that uses the photoelectric effect to generate or control a current.
- Photovoltaic cells and photoconductive cells are examples of photoelectric cells.
- Solar cells are a device that converts electromagnetic energy into electrical energy.
- Photovoltaic cells are used to power a wide variety of devices, and are often used in conjunction with a battery that stores the power they produce

COMPTON EFFECT:

Definition:

"The phenomenon in which a photon of frequency (ν) is scattered by an electron and the scattered photon has less frequency than it has before, is called Compton Effect."

Explanation:

Arthur Holly Compton (1892-1962) aimed short-wavelength light (i.e., x-rays or γ -rays) at various materials, and detected light scattered at various angles.

He observed that the scattered light had a slightly longer wavelength than did the incident light, and therefore a slightly lower frequency indicating a loss of energy. He explained this result based on the photon theory as incident photons colliding with electrons.

Particle Model of Photons with Energy and Frequency:

In Compton Effect experiment, there is an increase in wavelength of photons, due to their scattering by an electron. The impact results in one of the fundamentals of quantum mechanics, which represents wave and particle properties of light.

Compton successfully demonstrated that X-rays can be treated as discrete bundles, or quanta of electromagnetic energy. This concept was later termed "photon" by American physicist Gilbert Lewis.

Photons exhibit both particle-like properties, such as energy and momentum, and wave-like properties, such as frequency and diffraction.

The energy of a photon is dependent on its frequency, with lower energy photons having lower frequencies and longer wavelengths, as described by the equation:

$$E = \frac{h\nu}{\lambda}$$

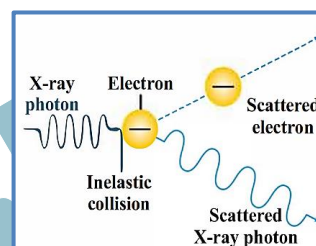
In Compton's experiment, photons collide with free or loosely bound electrons in matter. During these collisions, photons transfer part of their energy and momentum to the electrons, causing the electrons to recoil. This interaction results in the emission of new photons with lower energy and longer wavelengths, a phenomenon known as the Compton shift.

The shift in frequency of the scattered photons depends on the energy transferred to the electrons and is independent of the initial wavelength of the incident photons.

COMPTON EFFECT QUALITATIVELY:

Compton's Effect, as illustrated in Figure, can be qualitatively explained as follows:

1. **Initial Interaction:** A photon, which is a packet of electromagnetic energy, interacts with an electron in a material, typically a target like a metal or graphite.
2. **Scattering Process:** During the interaction, the photon transfers some of its energy and momentum to the electron. This transfer causes the photon to change its direction and wavelength (or equivalently, its frequency).
3. **Change in Photon Energy:** The scattered photon emerges with less energy (longer wavelength) than the initial incident photon.
4. The amount of energy lost by the photon is directly related to the energy and momentum gained by the electron.
5. **Quantum Nature:** Compton's Effect cannot be explained using classical wave theory alone. Classical wave theory predicts that light should scatter uniformly without a change in wavelength.
6. However, Compton's observations demonstrated that the scattered light has a shifted wavelength, indicating a particle-like interaction.



EXPERIMENTAL CONFIRMATION:

Compton conducted experiments where X-rays were targeted at graphite and the scattered X-rays were observed.

By measuring the scattering angle and change in wavelength, it is confirmed that the results were consistent with the predictions of quantum theory.

Wavelength Shift:

The wavelength shift observed in Compton scattering is directly proportional to the Compton wavelength, which depends on the mass of the electron and the Planck constant. This relationship provides crucial evidence for the particle nature of photons.

The Compton shift can be evolved as below:

$$\begin{aligned}\lambda_2 - \lambda_1 &= \frac{hc}{E_0} (1 - \cos\theta) \\ &= \frac{h}{m_0 c} (1 - \cos\theta) \\ \text{OR} \quad \Delta\lambda &= \frac{h}{m_0 c} (1 - \cos\theta)\end{aligned}$$

This equation describes the phenomenon known as Compton Effect.

$\Delta\lambda$ gives the change in photon wavelength due to scattering with a free electron and it is called **Compton shift**.

It's clear that Compton Shift is independent of the wavelength of the incident photon and depends on scattering angle.

The term $\lambda_c = \frac{h}{m_0 c} = 2.426 \times 10^{-12} \text{m}$, is called Compton Wavelength of the scattering particle i.e., electron

SELF-ASSESSMENT QUESTIONS:

1. Compare and contrast the photoelectric effect with the Compton effect and the Bohr model of the hydrogen atom in terms of their contributions to the development of quantum theory.

2. How does the Compton Effect demonstrate the particle nature of light?

WORKED EXAMPLE:

- 25.4: A beam of x-rays of wavelength 0.01 Å incident on a block of graphite. What is the wavelength of the x-rays scattered at an angle of 30°?

PAIR PRODUCTION:

Definition:

"The Phenomena in which photon energy is changed into an electron-positron pair, is called Pair Production".

Explanation:

Pair production is a phenomenon in quantum mechanics that occurs when a photon with sufficient energy interacts with matter and spontaneously transforms into a particle-antiparticle pair i.e., an electron and a positron.

A positron is a particle having mass and charge equal to that of electron but the charge being of opposite nature i.e. positive.

The creation of two particles with equal and opposite charges is essential for charge conservation in the universe. The positron is also known as an antiparticle of electron or anti-electron.

The interaction usually takes place in the electric field in the vicinity of a heavy nucleus (as shown in the figure) so that there is a particle to take up recoil energy and momentum is conserved.

In the process, radiant energy is converted into matter in accordance with Einstein's equation $E = mc^2$, and hence, is also known as materialization of energy.

For an electron or positron, the rest mass energy:

$$E_0 = m_0 c^2 = 0.51 \text{ MeV.}$$

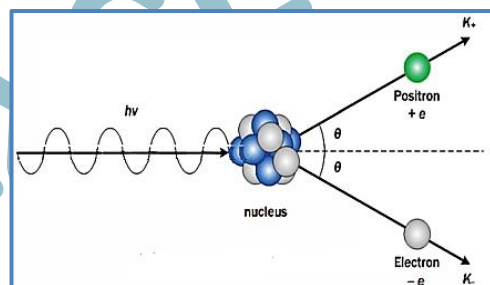
Thus, to create the two particles $2 \times 0.51 \text{ MeV}$ or 1.02 MeV energy is required.

For photons of energy greater than 1.02 MeV , the probability of pair production occurrence increases as the energy increases and the surplus energy is carried off by the two particles in the form of kinetic energy.

The process can be represented by the equation:

Energy of photon = Energy required for pair production + Kinetic energy of the particles

$$hf = 2m_0 c^2 + (K.E.)_{e^-} + (K.E.)_{e^+}$$



CONSERVATION LAWS OF PAIR-PRODUCTION:

For pair production to occur, a photon must have a minimum energy of 1.022 MeV .

This requirement arises because the rest mass of both (an electron and a positron) is 0.511 MeV each.

Therefore, the combined energy needed to create an electron-positron pair is $0.511 \text{ MeV} \times 2$ (1.022 MeV).

If the incident photon possesses energy exceeding 1.022 MeV , the surplus energy is distributed as kinetic energy between the electron and the positron.

Therefore, the pair production reaction is given as below:

$$\gamma \rightarrow e^- + e^+ \cong 1.02 \text{ MeV}$$

FACTS OF PAIR PRODUCTION PROCESS:

1. The pair production process obeys the law of conservation of energy, momentum and electric charge respectively.

2. During collision, the antiparticle of an electron, i.e., positron has the same physical properties as electron, except its charge, as both have opposite charge to each other. The sum of charges happens to be zero which is equal to photon before interaction. Therefore, the electric charge is conserved.
3. The law of conservation of energy is
 - (a) $hf = (K.E)_{e^-} + (K.E)_{e^+}$
 - (b) $hf = 2m_0c^2 + (K.E)_{e^-} + (K.E)_{e^+}$

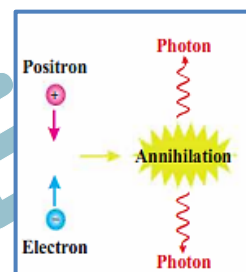
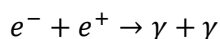
ANNIHILATION OF MATTER:

Electron-positron annihilation occurs when an electron and a positron collide. The result of the collision is the conversion of the electron and positron to the creation of gamma-ray photons.

CONSERVATION LAWS OF ANNIHILATION OF MATTER:

It is converse of pair production when a positron comes close to an electron they annihilate or destroy each other.

The matter of two particles changes into electromagnetic energy producing two photons in the γ -rays range.



Each photon has an energy equal to the rest mass of electrons 0.51 MeV. The two photons are produced moving in opposite direction in order to conserve momentum and energy which forbids the creation of a single gamma-ray, as shown in figure.

The charge is also conserved as net charge before and after is zero.

Pair production and Annihilation of matter supports that energy and mass are inter convertible i.e. $E=mc^2$.

WAVE NATURE OF PARTICLES:

Classical physics traditionally treats particles and waves as separate and distinct entities.

But the quantum theory proposed a dual character for radiation, suggesting that they could exhibit characteristics of both waves and particles.

The French physicist, Louis de Broglie proposed in 1924 that particles should also possess wavelike properties.

The wave nature of particles implies that particles can behave like waves, displaying properties such as reflection, refraction, interference, diffraction, and other wave-like characteristics.

This duality suggests that matter and radiation can co-exist both particle and wave attributes.

This dual behavior (both wave-like and particle like) is a fundamental principle of quantum mechanics, which helps us understand the behavior of matter and radiation at a tiny scale.

Examples:

Examples of particles that exhibit this dual nature include:

1. Matter particles, such as electrons, protons, and neutrons.
2. X-rays.
3. Photons.
4. Electromagnetic radiation.

Particles, like above, can behave like waves. This means they can show wave-like properties, like diffraction and interference.

DE BROGLIE WAVES:

Louis de Broglie's proposal of wave-particle duality for particles had a profound impact on the understanding of the nature of matter and laid the foundation for the development of quantum mechanics.

De Broglie's idea suggested that:

Both matter and energy could exhibit both particle and wave properties.

This unified description was a departure from the classical distinction between particles and waves, providing a more comprehensive understanding of the behavior of particles at the quantum level.

De Broglie's proposal was instrumental in the development of Niels Bohr's model of the atom. It helped to explain why electrons in quantized orbits do not radiate energy continuously but exist in stable orbits, as their orbits are analogous to standing waves.

DE BROGLIE WAVELENGTH:

According to de Broglie hypothesis:

"All matter particles like electrons, protons and neutrons in motion are associated with waves."

These waves are called de Broglie waves or matter waves.

Momentum of Photon:

According to Einstein's special theory of relativity:

$$E = mc^2 \text{ ----- (1)}$$

And according to Planck's assumption

$$E = hf \text{ ----- (2)}$$

By comparing equation (1) and (2), we get

$$mc^2 = hf$$

Divided both sides by c , we get

$$mc = \frac{hf}{c}$$

Since $mc = p$, the momentum. Therefore

$$p = \frac{hf}{c}$$

By putting the value of $c = \lambda f$, we get

$$p = \frac{hf}{\lambda f}$$

$$p = \frac{h}{\lambda}$$

Wavelength of photon:

The wavelength of a photon in terms of its momentum is

$$\lambda = \frac{h}{p}$$

but $p = mv$, then

According to de Broglie, the above equation is completely applicable to matter particles as well. Therefore, for a particle of mass m travelling with speed v , the wavelength is given by

$$\lambda = \frac{h}{mv}$$

This wavelength of the matter waves is known as de Broglie wavelength. This equation relates to the wave character (the **wavelength**) and the particle character (the **momentum**) through Planck's constant.

DAVISSON - GERMER EXPERIMENT:

De Broglie hypothesis of matter waves was experimentally confirmed by Clinton Davisson and Lester Germer in 1927.

They demonstrated that electron beams are diffracted when they fall on crystalline solids.

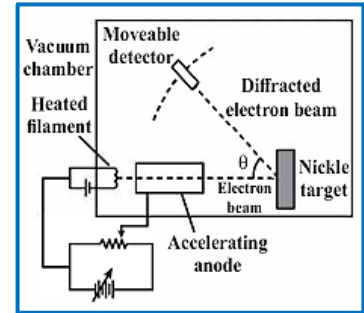
Since crystals can act as a three-dimensional diffraction grating for matter waves, therefore electron waves incident on crystals are diffracted off in certain specific directions.

Figure shows a schematic representation of the apparatus for the experiment.

CONSTRUCTION:

The experimental parts are given as under:

1. An electron gun emits electrons via thermionic emission produced by the tungsten filament used in it, i.e., when heated to a specific temperature.
2. Two opposite charged plates known as Electrostatic Particle Accelerator, which accelerates the electrons at a certain potential.
3. The accelerator is enclosed within a cylinder called Collimator which is a narrow passage for the electrons along its axis.
4. The target is a Nickel crystal on which the electron beam is fired normally.
5. When the electrons are scattered from Ni crystal, these are captured by the semicircular movable detector.



DAVISSON GERMER EXPERIMENT AND DE BROGLIE RELATION:

Let us consider an electron of mass (m_0), charge (e) accelerated from rest through a potential (V).

Expression for Momentum:

Then, the kinetic energy $K.E$ of the electron is equal to the work done (In eV) on it by the electric field:

$$K.E = eV$$

$$\text{Now, } KE = \frac{1}{2} m_0 v^2 = \frac{p^2}{2m_0}.$$

Therefore

$$\begin{aligned} \frac{1}{2} m_0 v^2 &= Ve \\ m_0 v^2 &= 2Ve \dots\dots\dots (1) \end{aligned}$$

Multiply both sides of equation by " m_0 ", we have

$$\begin{aligned} m_0^2 v^2 &= 2m_0 Ve \\ m_0 v &= \sqrt{2m_0 Ve} \\ \text{OR } p &= \sqrt{2m_0 Ve} \dots\dots\dots (2) \end{aligned}$$

De Broglie Wavelength:

The de Broglie wavelength of electron is given by

$$\lambda = \frac{h}{p} \dots\dots\dots (3)$$

By putting the value of " p " from equation (2) in equation (3), we get

$$\lambda = \frac{h}{\sqrt{2m_0 Ve}}$$

By using $h = 6.626 \times 10^{-34}$ J-s, $m_0 = 9.11 \times 10^{-31}$ kg, and $e = 1.6 \times 10^{-19}$ C, we have

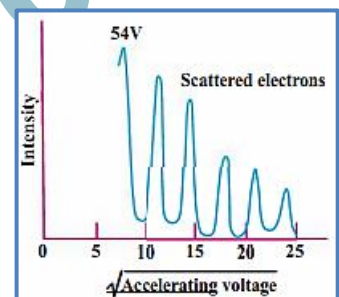
$$\lambda = \frac{1.227}{\sqrt{V}} \text{ nm}$$

Where " V " is the magnitude of accelerating potential in volts.

As de Broglie wavelength is associated with electrons for $V = 54$ V from graph, then we have $\lambda = 0.167$ nm

Conclusion:

This experiment confirms the wave nature of electrons and the de Broglie relation.



ELECTRON MICROSCOPE:

Introduction:

Ernst Ruska (1906-1988), a German engineer and academic professor, built the first Electron Microscope (EM) in 1931, and the same principles behind his prototype still govern modern EMs.

Definition:

A special type of microscope that uses an electron beam with its wavelike properties to illuminate a specimen (nanometers) able to magnify objects in high resolution, which are formed by controlled use of electrons in a vacuum captured on a phosphorescent screen.

Explanation:

Electron microscopes make practical use of the wave nature of electrons, which is thousands of times shorter than visible light which enables the electron microscope to distinguish details that are not visible with optical microscopes.

In the electron microscope design, the wave properties of electrons have been utilized with higher resolution, which is giving a great improvement in visualization.

TYPES OF ELECTRON MICROSCOPE:

There are two types of electron microscopes which are:

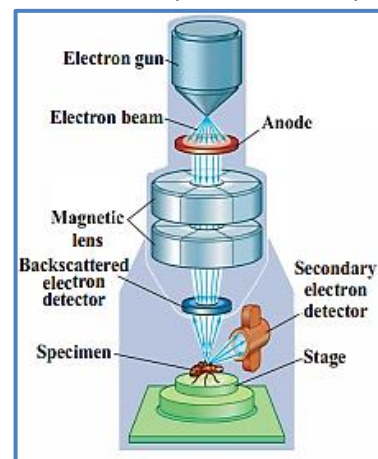
1. Transmission electron microscope (TEM), which produces a two-dimensional image, and
2. Scanning electron microscope (SEM), which produces images with a three-dimensional

3. BASIC PRINCIPLE OF ELECTRON MICROSCOPE:

4. De Broglie hypothesis is used in the field of electron optics.
5. De Broglie hypothesis of electron have wave paved the way for the development of the electron microscope (EM), which can produce images of much greater magnification than an optical microscope.

CONSTRUCTION AND WORKING:

- In both types, i.e., TEM & SEM, the objective and eyepiece lenses are actually magnetic fields that exert force on the electrons to bring them to a focus.
- The fields are produced by carefully designed current-carrying coils of wire.
- EM's measure the intensity of electrons, producing mono chromatic image. Color is often added artificially to highlight.
- The probe tip of scanning electron microscope, as it is moved horizontally, automatically moves up and down to maintain a constant tunneling current, and this motion is translated into an image of the surface.



APPLICATIONS:

- Electron microscopes are used to investigate the ultrastructure of a wide range of biological and inorganic specimens including microorganisms, cells, large molecules, biopsy samples, metals, and crystals.
- Industrially, electron microscopes are often used for quality control and failure analysis.
- Modern electron microscopes produce electron micrographs using specialized digital cameras and frame grabbers to capture the images.

- The advancement of microbiology is significantly indebted to the electron microscope, which has revolutionized our understanding of microorganisms such as bacteria, viruses, and other pathogens,
- thereby greatly enhances the effectiveness of disease treatments.

ROLE OF ELECTRON MICROSCOPE:

1. Higher Resolution:

- Electron microscopes offer significantly higher resolution compared to optical microscopes.
- It allows observation at the nanometer scale due to the shorter wavelength of electrons.

2. Transmission Electron Microscope (TEM):

- TEMs study internal structures of thin specimens, producing detailed images of cells, organelles, and crystalline structures.
- Valuable in biology, materials science, and nanotechnology.

3. Scanning Electron Microscope (SEM):

- SEMs provide 3D surface images by scanning specimens and detecting emitted secondary electrons.
- Widely used in biology, geology, and materials science for surface analysis.

4. Energy- Dispersive X-ray Spectroscopy (EDS):

With EDS detectors, electron microscopes analyze elemental composition by detecting X-rays emitted when high energy electrons interact with the sample.

5. Materials Science:

It is crucial for studying the microstructure of materials, aiding understanding of relationships between microstructure and properties in metals, ceramics, and polymers.

6. Nanotechnology:

Essential for imaging and characterizing nanomaterials, contributing to the development of new materials and devices through observation and manipulation of nanoparticles and nanostructures.

7. Advancements in Medicine:

This contributes to medical research by providing insights into the structure of viruses, bacteria, and cellular organelles, aiding understanding of diseases and therapeutic development.

8. Quality Control in Industry:

It is used across industries for quality control and failure analysis, helping identify defects, analyze material composition, and ensure product integrity at the microscopic level.

UNCERTAINTY PRINCIPLE:

Introduction:

In 1927, Werner Heisenberg suggested a principle that applies to measuring the properties of quantum-sized objects (e.g., atomic and sub-atomic particles).

HEISENBERG UNCERTAINTY PRINCIPLE:

Statement:

"It is impossible to measure simultaneously certain pairs of properties of a subatomic particle, such as its position and momentum or energy and time, with absolute certainty". OR

“The product of the uncertainty Δx in the position of a particle at some instant and the uncertainty Δp in its momentum at the same instant is approximately equal to Plank’s constant h ”.

Mathematically:

Mathematically it can be written as: $\Delta x. \Delta p \approx h$

Explanation:

According to Heisenberg Uncertainty Principle, it is impossible to measure simultaneously certain pairs of properties of a subatomic particle, such as its position and momentum, with absolute certainty.

In other words, the more precisely we know the position of a particle, the less precisely we be able to know its momentum, and vice versa.

This principle develops from the wave-particle duality of particles in quantum physics, where particles can exhibit both wave-like and particle-like behaviors.

The formal inequality relating to the standard deviation of position Δx and the standard deviation of momentum Δp are given here:

$$\begin{aligned}\Delta x. \Delta p &\geq \frac{h}{4\pi} \\ \Delta x. \Delta p &\geq \frac{h/2\pi}{2} \\ \Delta x. \Delta p &\geq \frac{\hbar}{2}\end{aligned}$$

Where $\hbar$ (pronounced "h-bar") is the reduced Planck constant.

The Heisenberg uncertainty principle can also be applied to other pairs of complementary quantum properties, such as energy and time.

$$\Delta E. \Delta t \geq h/2\pi$$

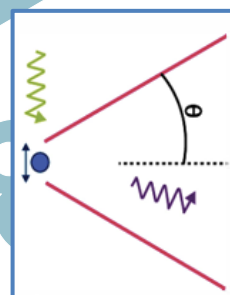
Here, ΔE and Δt represents the uncertainties in energy and time respectively.

The Uncertainty Principle has profound consequences for understanding quantum world. It means that at the subatomic level, there are inherent limits to how precisely we can measure certain properties of particles. This is not due to restrictions on our measuring instruments, but it is a fundamental aspect of the quantum nature of the universe.

Applications:

The uncertainty principle mainly applied to experiments in physics labs, however some real-world effects are given.

For example, in the sun's core, it is the uncertainty in the position of two hydrogen nuclei (protons) that allows for there to be a chance that they will overlap and fuse together. This fusing releases a tremendous amount of energy, which we receive as light. A pretty significant effect on our everyday lives.



SUMMARY

- A black body absorbs all incident radiation and releases it in a continuous spectrum of colors, depending upon its temperature.
- The energy per unit volume per unit wavelength of black body radiation is directly proportional to the square of the wavelength and increases indefinitely as the wavelength becomes smaller.
- Different atoms and molecules can emit or absorb energy in discrete quantities only.
- When electromagnetic radiation like light is shined on certain metallic materials, electrons are emitted due to absorption of light by the electrons on the surface of material.
- Work function is the minimum energy that must be supplied to the electron for it to leave the metal.

- For each substance there exists a characteristic frequency ν_0 such that for radiation with frequency below the photoelectrons are not ejected from the surface.
- Compton scattering refers to X-ray or gamma-ray photons undergo a change in wavelength and momentum when they collide with charged particles, typically electrons.
- Change in wavelength gives the change in photon wavelength due to scattering with a free electron.
- When the photon interacts with the strong electric field around the nucleus, it undergoes a change of state and is transformed into two particles (creating matter from energy):
- Electron-positron annihilation occurs when an electron and a positron (the antiparticle of the electron) collide.
- All matter particles like electrons, protons and neutrons in motion are associated with waves.
- A microscope which uses a beam of accelerated electrons as a source of lighting.
- It is impossible to measure simultaneously all quantum properties of a particle with equal accuracy

TEXTBOOK EXERCISE:

SECTION- A: MULTIPLE CHOICE QUESTIONS (MCQS)

- If h is the Planck constant, then h is:
 (a) $2\pi h$ (b) $2h$ (c) $h/2$ (d) $h/2\pi$ (e) $2h/\pi$
- In a photoelectric effect experiment the stopping potential is:
 (a) the energy required to remove an electron from the sample
 (b) the kinetic energy of the most energetic electron ejected
 (c) the potential energy of the most energetic electron ejected
 (d) the electric potential that causes the electron current to vanish
- The work function for a certain sample is 2.3 eV. The stopping potential for electrons ejected from the sample by 7.0×10^{14} -Hz electromagnetic radiation is: (2024)
 (a) 0 (b) 0.60 V (c) 2.3V (d) 2.9V (e) 5.2V
- In Compton scattering from stationary particles the maximum change in wavelength can be made smaller by using:
 (a) higher frequency radiation (b) lower frequency radiation
 (c) more massive particles (d) less massive particles
- Of the following, Compton scattering from electrons is most easily observed for: (MP:25)
 (a) Infrared light (b) visible light (c) ultraviolet light (d) X-rays
- In the Compton scattering from stationary electrons the largest change in wavelength occurs when the photon is scattered through
 (a) 0° (b) 45° (c) 90° (d) 180°
- A free electron has a momentum of $5.0 \times 10^{-24} \text{ kg} \cdot \text{m/s}$. The wavelength of its wave function is: (MP:25)
 (a) $1.3 \times 10^{-8} \text{ m}$ (b) $1.3 \times 10^{-10} \text{ m}$ (c) $2.1 \times 10^{-11} \text{ m}$ (d) $2.1 \times 10^{-13} \text{ m}$
- In Photoelectric Effect, when light falls on the surface of metal, the material should emit
 (a) Electrons (b) Protons (c) Positrons (d) Neutrons
- Threshold frequency is defined as the Frequency of incident light which can cause photo electric emission.
 (a) Maximum (b) Minimum (c) Average (d) highest
- The amount of energy which is necessary to start photo electric emission is called:
 (a) Maximum (b) Average (c) Minimum (d) Littlest

SECTION- B: SHORT ANSWERED QUESTIONS

1. Differentiate between wave and particle. (2024)
2. Is it possible for the de Broglie wavelength of a particle?
3. Estimated Broglie wavelength of a cricket ball on the pitch?
4. Differentiate between the continuous and discrete emission of radiation?
5. What is threshold frequency?
6. How has the photoelectric effect been applied in real-world technologies or devices, and what are its practical implications?
7. What are the advantages of an electron microscope over an optical microscope?
8. Is it possible to create only an electron through matter and photon interaction?
9. Give construction of electron microscope?
10. Elaborate on the particle nature of electromagnetic radiation?

SECTION- C: LONG ANSWERED QUESTIONS

1. Describe how energy is distributed over the wavelength range for different temperatures.
2. Explain the particle model of light in terms of photons with particular energy and frequency. (2024)
3. Describe conservation laws in pair production and annihilation of matter.
4. Describe Compton's effect qualitatively.
5. Explain how the very short wavelength of electrons, and the ability to use electrons and magnetic fields to focus on them, allows electron microscope to achieve very high resolution.
6. Describe the impact of de Broglie proposal that any kind of particle has both wave and particle properties.
7. Describe the confirmation of de Broglie proposal by Davisson and Germer experiment in which the diffraction of electrons by the surface layers of a crystal lattice was observed.

SECTION- D: NUMERICAL

1. The Sun's surface temperature is 5700 K. (i) How much power is radiated by the Sun? (ii) Given that the distance to Earth is about 200 Sun radii, what is the maximum power possible from one square kilometer solar energy installation? (iii) What is the wavelength of maximum intensity of solar radiation? (5.98x10⁷ W/m², 3.6x10²⁶ Watts, 5.1x10⁻⁷ m)
2. The temperature of your skin is approximately 32°C. What is the wavelength at which the peak occurs in the radiation emitted from your skin? (9.05x 10⁻⁵ m)
3. An FM radio transmitter has a power output of 100 kW and operates at a frequency of 94 MHz, How many photons per second does the transmitter emit? (6.23x10⁻²⁶ J, 1.61x10³⁰ s⁻¹)
4. A light source of wavelength illuminates a metal and ejects photoelectrons with a maximum kinetic energy of 1.0 eV. A second light source with half the wavelength of the first ejects photoelectrons with a maximum kinetic energy of 4.0 eV. Determine the work function of the metal. (2 eV)
5. A 430 nm violet light is an incident on a calcium photo electrode with a work function of 2.71 eV. Find the energy of the incident photons and the maximum kinetic energy of ejected electrons. (2, 88 eV, 0.17 eV)
6. Cut-off frequency for the photoelectric effect in some materials is 8x10¹³ Hz. When the incident light has a frequency of 1.2x 10¹⁴ Hz, the stopping potential is measured as - 0.16 V. Estimate a value of Planck's constant from these data and determine the percentage error of your estimation. (5.17 eV, 1.30 V)
7. The work function of some metals is listed below. The number of metals which will show photoelectric effect when light of 300 nm wavelength falls on the metal is: (4 Metals)

Metal	Li	Na	K	Mg	Cu	Ag	Fe	Pt	W
ϕ in eV	2.4	2.3	2.2	3.7	4.8	4.3	4.7	6.3	4.75

8. X-rays with an energy of 300 keV undergo Compton scattering with a target. If the scattered X-rays are detected at 30° relative to the incident X-rays, determine the Compton shift at this angle, the energy of the scattered X-rays, and the energy of the recoiling electron. **(0.35 pm, 278keV, 22keV)**
9. A photon with a wavelength of 6.0×10^{-12} m collides with an electron. After the collision the photon wavelength is found to have been changed by exactly one Compton Wavelength is 2.43×10^{-12} m).
 - (i) What is the photon's wavelength after collision? **(MP:25)**
 - (ii) Through what angle has been deflected in this collision?
 - (iii) What is the angle for the electron after the collision?
 - (iv) What is the electron's kinetic energy, in eV, after collision? **(8.4×10^{-12} m, 90° , $< 90^\circ$ 5.9×10^4 eV)**
10. Find the de Broglie wavelength of an electron in the ground state of hydrogen. **(3.324 Å)**
11. Determine the minimum uncertainties in the positions of the following objects if their speeds are known with a precision of 1.0×10^{-3} m/s: (a) an electron and (b) a bowling ball of mass 6.0 kg. **(5.8cm, 33 m)**

ASSIGNMENT: 25

SECTION- A: MULTIPLE CHOICE QUESTIONS

Black Body Radiation:

1. The energy radiated from black body is directly proportional to the fourth power of absolute temperature' This law is: **(2023)**
 - (a) Wein's Law
 - (b) Rayleigh Jeans Law
 - (c) Stefan's Law.
 - (d) Plank's Law
2. Stefan Boltzmann law is: **(2019)**
 - (a) $E = \alpha T$
 - (b) $E = \alpha T^4$
 - (c) $E = \alpha T^3$
 - (d) $E = \alpha T^2$
3. 1 MeV, is equal to: **(2015)**
 - (a) 1.6×10^{-19} J
 - (b) 1.6×10^{-13} J**
 - (c) 1.6×10^{18} J
 - (d) 1.6×10^{19} J
4. One atomic mass unit is equal to: **(2005)**
 - (a) 1.6×10^{-19} J
 - (b) 1×10^{-27} kg
 - (c) 931×10^6 eV**
 - (d) 9×10^9 eV
5. Stefan Boltzmann's law is: **(2016)**
 - (a) $E = \sigma T$
 - (b) $E = \sigma T^2$
 - (c) $E = \sigma T^3$
 - (d) $E = \sigma T^4$**
6. The mathematical expression $\lambda_{\max} \times T = \text{constant}$ is called: **(2014, 12, 08)**
 - (a) Stefan's law
 - (b) Rayleigh-Jeans law
 - (c) Planck's law
 - (d) Wien's displacement law**
7. The energy radiated per second per unit area from the surface of a black body is directly proportional to its absolute temperature raised to power: **(2014)**
 - (a) One
 - (b) Two
 - (c) Three
 - (d) Four**
8. As the temperature of Black Body rises, the wavelength corresponding to the maximum intensity shifts towards: **(2013, 11)**
 - (a) Similar wavelength
 - (b) Shorter wavelength**
 - (c) Longer wavelength
 - (d) none of these
9. A perfect black body: **(2008)**
 - (a) is a perfect absorber of radiation
 - (b) has a unit absorptive power
 - (c) is the most efficient radiator
 - (d) All of them**
10. The radiation from a black body depends upon the: **(2005)**
 - (a) Material of the body
 - (b) Size of the body
 - (c) Shape of the body
 - (d) Temperature of the body**
11. Max Planck is well known because of his: **(2002E)**
 - (a) Energy equation**
 - (b) Energy Conservation
 - (c) Wave Particle Duality
 - (d) Momentum Conversation

12. The absorption of the incident radiation by a perfect black body is: (2002E)
 (a) 0% (b) 100% (c) 90% (d) 50%
13. As the temperature of a black body is raised, the wavelength corresponds to the maximum intensity:
 (a) Shift towards longer wavelength (b) Shifts towards shorter wavelength (2001)
 (c) remains the same (d) None of the above

Photon:

14. Rest mass of photon is: (2023)
 (a) Infinite (b) Zero (c) 9.11×10^{-31} kg (d) 1.67×10^{-27} kg
15. This moves with speed of light: (2021)
 (a) photon (b) proton (c) positron (d) electron
16. This particle has no charge, no rest mass and can interact with all charged as well as neutral particles: (2018)
 (a) Alpha particle (b) Neutron (c) Photon (d) Positron
17. The rest mass of a photon is: (2016, 15, 13)
 (a) 1 (b) -1 (c) zero (d) infinite
18. The formula of the momentum of a photon is. (2006) (MP:26)
 (a) $\frac{hc}{\lambda}$ (b) $\frac{h}{\lambda}$ (c) $\frac{\lambda}{h}$ (d) $\frac{c}{\lambda}$

Photoelectric Effect:

19. In Photoelectric Effect, increasing the intensity of light will increase the: (MP:26)
 20. (a) Frequency of light (b) Wavelength of light (c) Number of photons (d) Number of photoelectrons
21. Least amount of energy required to emit an electron from surface of metal, in photoelectric effect is called: (2025) (MP:26)
 (a) Threshold frequency (b) Binding energy (c) Work function (d) Stopping potential
22. Photoelectric effect proves this nature of light: (2022)
 (a) Wave (b) Particle (c) Polarization (d) Diffraction
23. In order to increase the number of electrons in photoelectric effect, this should be increased: (2021)
 (a) Intensity of light (b) Threshold frequency (c) Work function (d) Wavelength of light
24. In the phenomenon of photoelectric effect, if the frequency of incident radiation is increased, the required stopping potential: (2017)
 (a) Increases (b) decreases (c) does not change (d) become zero
25. A photoelectric cell transforms light energy into: (2011)
 (a) Heat energy (b) Magnetic energy (c) Electrical Energy (d) Sound energy
26. In order to increase the kinetic energy an ejected photoelectron there should be an increase in: (2011)
 (a) Frequency of Radiation (b) Wavelength of Radiation
 (c) Intensity of Radiation (d) Both Wavelength and intensity
27. The photoelectric emission takes place if: (2009)
 (a) $h\nu < \phi_0$ (b) $h\nu > \phi_0$ (c) $h\nu_0 > \phi$ (d) $V_0 < \phi_0$
28. The frequency of the incident radiation corresponding to the work function is called: (2005)
 (a) Fundamental frequency (b) Working frequency
 (c) Critical frequency (d) Threshold frequency
29. The Einstein Photo –Electric Equation is: (2002E)
 (a) $V_0 e = h\nu + \frac{1}{2} mv^2$ (b) $h\nu = mc^2 + V_0$ (c) $V_0 e + h\nu - h\nu_0$ (d) $V_0 e = [h\{1/\lambda - 1/\lambda_0\}]$
30. The maximum kinetic energy of photoelectron emitted from a metal depends upon. (2002M)

- (a) The frequency of incident light only (b) The wavelength of incident light only
(c) Work function of the metal only (d) All of them

Compton Effect:

31. Compton effect proves the: **(2023) (MP:26)**
(a) Photon Theory of Light (b) Dual Nature of Light (c) Nature of Light (d) Uncertain Nature of Light
32. This quantity is called Compton's wavelength: **(2021)**
(a) h/m_0c^2 (b) hc/m_0 **(c) h/m_0c** (d) m_0c/h
33. In Compton's scattering experiment, the scattered photon has a: **(2015)**
(a) Frequency less than that of incident photon (b) frequency greater than that of incident photon
(c) Same frequency as that of incident photon (d) wavelength shorter than that of incident photon
34. In Compton Effect the scattered photon has: **(2007)**
(a) Greater frequency, smaller wavelength (b) Greater frequency, greater wavelength
(c) Smaller frequency, greater wavelength (d) None of them
35. As a result of elastic collision between a photon and an electron the: **(2005)**
(a) Wavelength of the photon is increased (b) Frequency of the photon is increased
(c) Energy of the photon is increased (d) X-rays are produced.

Pair Production:

36. In this process, matter is converted into energy: **(2022)**
(a) Pair production (c) Compton effect (c) Photo electric effect (d) Annihilation of matter
37. In pair production the electron and positron move in opposite direction to conserve: **(2022)**
(a) Momentum (b) Charge (c) Energy (d) Mass
38. The minimum energy required for pair production is: **(2022)**
(a) 2.01 MeV (b) 1.02 MeV (c) 0.21 MeV (d) 1.20 MeV
39. The minimum energy of photon, required for a pair production is: **(2021)**
(a) 1.02 keV (b) 10.2 keV (c) 102 keV (d) 1020 keV
40. A positron is an anti-particle of: **(2021)**
(a) proton **(b) electron** (c) neutron (d) photon
41. If m_0 is the rest mass of an electron. Pair production takes place only if the minimum energy of incident photon is equal to: **(2019)**
(a) $\frac{2m_0}{c^2}$ (b) $\frac{2}{m_0c^2}$ (c) m_0c^2 (d) $2 m_0c^2$
42. The phenomenon of pair production takes place if the energy of photon is greater than: **(2010)**
(a) 1.0MeV **(b) 1.02MeV** (c) 0.051MeV (d) None of these
43. The reverse of pair production is known as: **(2003)**
(a) Fusion (b) Fission
(c) Annihilation of Electrons and positron. (d) Annihilation of Electron and protons
44. Pair production is only possible when incident photon has the wavelength of the order of: **(2002E)**
(a) 10^{-4}m (b) 10^{-6}m (c) 10^{-10}m (d) 10^{-14}m

De Broglie's Waves:

45. An electron of mass 'm' and charge 'e' is accelerated by a difference of potential V, its speed will be:
(a) $v = \sqrt{\frac{2Ve}{m}}$ (b) $v = \sqrt{\frac{2Vm}{e}}$ (c) $v = \sqrt{2Vme}$ (d) $v = \sqrt{\frac{m}{2Ve}}$ **(2022)**
46. de-Broglie wavelength is: **(2018, 14, 11)**

- (a) $\lambda = \frac{mv}{h}$ (b) $\lambda = \frac{h}{mv^2}$ (c) $\lambda = \frac{h}{mv}$ (d) $\lambda = \frac{mh}{v}$
47. The wavelength of a material particle of mass 'm' moving with a velocity 'v' is given by: (2007)
- (a) $\lambda = \frac{hv}{m}$ (b) $\lambda = \frac{m}{hv}$ (c) $\lambda = \frac{m}{hv}$ (d) $\lambda = \frac{h}{mv}$
48. According to uncertainty principle: (2012)
- (a) $(\Delta x)(\Delta t) = \hbar$ (b) $(\Delta x)(\Delta p) = \hbar$ (c) $(\Delta E)(\Delta p) = \hbar$ (d) $(\Delta x)(\Delta E) = \hbar$
49. According to the principle of Uncertainty: (2009)
- (a) $\Delta E/\Delta t \approx h$ (b) $\Delta E \approx \Delta t/h$ (c) $(\Delta E)(h) \approx \Delta t$ (d) $(\Delta E)(\Delta t) \approx h$

SECTION- B: SHORT QUESTIONS

Black Body Radiation:

1. State the laws governing black body radiation. Also write their mathematical expressions. (2025)

Photoelectric Effect:

2. Define photo-electric effect. Explain it on the basis of the quantum theory of light and obtain Einstein's photo-electric equation. (2001)
3. What is photoelectric effect? What is meant by the terms? (i) threshold frequency (ii) work function (iii) saturation current (iv) stopping potential (2019)

Compton Effect:

4. Explain Compton Effect and obtain an expression for the increase in the wavelength of the scattered photon. (2022)

Pair Production:

5. Define Pair Production and Pair Annihilation. (MP:26)
6. Define Annihilation of matter. What do you understand by pair Production? Describe the condition for pair production with the help of an equation.

De Broglie's Waves:

7. Describe De Broglie Hypothesis for the description of wave nature of particles and prove it experimentally.

Uncertainty Principle:

8. State and explain Heisenberg's Uncertainty Principle and derive the relative mathematical expression.

NUMERICAL PROBLEMS

Photoelectric Effect:

1. In photo electric experiment, a stopping potential of 1.25 volt is needed. Find the maximum kinetic energy and maximum speed of photo electron. $e = 1.6 \times 10^{-19} \text{C}$; $m_e = 9.1 \times 10^{-31} \text{kg}$ (2022)
2. Sodium surface is shined with light of wavelength $3 \times 10^{-7} \text{m}$. If the work function of Na = 2.46 eV, find the K.E of the Photoelectrons and also the cut off wavelength. (Ans: 1.68 eV, $5.05 \times 10^{-7} \text{m}$) (2021, 16, 12, 09, 03E, 03M)
3. When light of the wavelength $4000 \text{\AA}$ falls on a metal surface stopping potential is 0.6volt; find the value of the work function of the metal; $h = 6.625 \times 10^{-34} \text{J-S}$. (Ans: 2.5 eV) (2001)
4. The range of visible light is $4000 \text{\AA}$ to $7000 \text{\AA}$. Will photoelectrons be emitted by a copper surface of work function 4.4 eV, when illuminated by visible light? Give mathematical proof of your answer. (Ans: $E_1 \text{ \& } E_2 < \phi$, No photoelectron emission) (1998)
5. The work function of a metal is 2 eV. The light of wavelength $3000 \text{\AA}$ is made to fall on it. Find the kinetic energy of the fastest emitted photoelectron. $C = 3 \times 10^8 \text{m/s}$, $h = 6.63 \times 10^{-34} \text{J.S}$. (Ans: 2.14 eV) (1994)

Compton Effect:

6. X-rays of wavelength 3.64×10^{-10} m are used in Compton scattering process. Find the fractional change in wavelength for a scattering angle of 120° . (Give $h = 6.63 \times 10^{-34}$ Js, $m_0 = 9.1 \times 10^{-31}$ kg, $c = 3 \times 10^8$ m/s)
(Ans: 3.642×10^{-10} m) (2017, 07) (2023)

De Broglie's Waves:

7. An electron is accelerated through a potential difference of 50 volt. Calculate its deBroglie wavelength ($h = 6.63 \times 10^{-34}$ J.s; $e = 1.6 \times 10^{-19}$ C; $m = 9.11 \times 10^{-31}$.kg)
8. If the electron beam in a TV picture tube is accelerated by 10 K volt. What will be the de Broglie wavelength of an electron? ($h = 6.63 \times 10^{-34}$ JS, $e = 1.6 \times 10^{-19}$ C, $m = 9.11 \times 10^{-31}$ kg) (Ans: 1.23×10^{-11} m)

Uncertainty Principle:

9. An electron exists within a region of 10^{-10} m, find its momentum uncertainly and approximate kinetic energy. ($h = 1.05 \times 10^{-34}$ j-sec, $m = 9.1 \times 10^{-31}$ kg) (Ans: 1.05×10^{-24} N.s, 3.78 eV) (2006)

Pair Production:

10. A photon of wavelength $0.004 \text{ \AA}$ in the vicinity of a heavy nucleus produces an electron-positron pair in MeV, if the kinetic energy of positron is twice that of electron.
($m_0 c^2 = 8.19 \times 10^{-14}$ joules, $h = 6.63 \times 10^{-34}$ J.s and $c = 3 \times 10^8$ m/s) (2018)
11. Pair annihilation occurred due to a head on collision of an electron and positron having the same kinetic energy, producing pair of photons each having energy of 2.5 MeV. What were there kinetic energies before collision? Given $m_0 c^2 = 0.511$ MeV. (2013)

SECTION- C: DETAILED ANSWERED QUESTIONS:

12. What is photoelectric effect? Define the terms Threshold frequency, stopping potential and work function. Also derive Einstein's photoelectric equation. (MP:26)
13. What is Photoelectric Effect? Explain its important result. Derive Einstein's photoelectric equation. (2023)